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Calculation

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The mathematical model is

$$\begin{aligned}\frac{dV}{dt} &= \frac{I}{C}, & V(0) &= 0, \\ \frac{dI}{dt} &= \frac{-V}{L} - \frac{RI}{L} + \frac{V_B}{L}, & I(0) &= 0,\end{aligned}$$

where $V_B = 5\text{ V}$, $R = 0.1\ \Omega$, $L = 4 \times 10^{-9}\text{ H}$, and $C = 0.5 \times 10^{-9}\text{ F}$.

Exercise A

2/2 points (graded)

The system of differential equations has one equilibrium point $(V_{\text{eq}}, I_{\text{eq}})$. Calculate V_{eq} and I_{eq} exactly.

The value of V_{eq} is

✓ Answer: 5

The value of I_{eq} is

✓ Answer: 0

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Exercise B

1/1 point (graded)

Linearise the differential equations around the equilibrium point (V_{eq}, I_{eq}) . That should not be too difficult!

The characteristic equation for the eigenvalues of the Jacobian matrix $J \left(\begin{bmatrix} V_{eq} \\ I_{eq} \end{bmatrix} \right)$ is

☒ $\lambda^2 + \frac{R}{L}\lambda + \frac{1}{LC} = 0$ ✓

☐ $\lambda^2 - \frac{R}{L}\lambda + \frac{1}{LC} = 0$

☐ $\lambda^2 + \frac{R}{L}\lambda - \frac{1}{LC} = 0$

☐ $\lambda^2 - \frac{R}{L}\lambda - \frac{1}{LC} = 0$

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i Answers are displayed within the problem

Exercise C

1/1 point (graded)

The eigenvalues of the Jacobian matrix $J \left(\begin{bmatrix} V_{eq} \\ I_{eq} \end{bmatrix} \right)$ are approximately

☐ $\lambda \approx \left(-\frac{1}{80} \pm \frac{\sqrt{2}}{2} \right) \times 10^9$

☐ $\lambda \approx \left(+\frac{1}{80} \pm \frac{\sqrt{2}}{2} \right) \times 10^9$

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☒ $\lambda \approx \left(-\frac{1}{80} \pm \frac{\sqrt{2}}{2}i \right) \times 10^9$ ✓

☐ $\lambda \approx \left(+\frac{1}{80} \pm \frac{\sqrt{2}}{2}i \right) \times 10^9$

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Exercise D

1/1 point (graded)

Determine the type of equilibrium point (V_{eq}, I_{eq}) .

(V_{eq}, I_{eq}) is a

☐ Stable node

☐ Unstable node

☐ Saddle point

☒ Stable spiral point ✓

☐ Unstable spiral point

☐ Circle point

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When the eigenvalues of the Jacobian matrix are complex numbers: $\lambda = a \pm bi$, the solutions for $\vec{M}(t)$ contain factors $e^{at} \cos(bt)$ and $e^{at} \sin(bt)$.

The radial frequency ω , the frequency f , and the period T are related by $T = \frac{1}{f} = \frac{2\pi}{\omega}$.

Exercise E

1/1 point (graded)

In this case, where $R = 0.1 \Omega$, $L = 4 \times 10^{-9} \text{H}$, and $C = 0.5 \times 10^{-9} \text{F}$, calculate the length of each cycle in the solution in nanoseconds (10^{-9}s).

$T \approx$

Give your answer in nanoseconds and with two decimals.

✓ Answer: 8.89

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You have used 1 of 3 attempts

ⓘ Answers are displayed within the problem

Exercise F

1/1 point (graded)

Now keep $R = 0.1 \Omega$ and $L = 4 \times 10^{-9} \text{H}$, but determine the capacitance and C such that $f = 100 \text{MHz}$ and $T = 10 \times 10^{-9} \text{s}$. Give your answer in nanofarad (10^{-9}F).

$C \approx$

Give your answer in nanofarad and with three decimals.

✓ Answer: 0.633

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ⓘ Answers are displayed within the problem

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In the last exercises, you have seen that you can predict the cycle length with an analytical calculation and that you can use that to design your system. On the next page you will investigate again what happens when the resistance of the circuit is changed, but now from a theoretical point of view.

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English ▼

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